

Technical Document #928: Data Engineering - Comprehensive Overview

Technical Document #928: Data Engineering - Comprehensive Overview

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Data lakes store raw data in its native format.
- Stream processing handles continuous data streams in real-time.
- Data lineage tracks the origin and movement of data through systems.